ChaoWei Liao

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Experience

Anduril Industries Los Angeles Metro Area, CA, USA

Senior Manufacturing Test Engineer Mar. 2026 - Present

- Own end-of-line test and calibration for a 16-station production line (4 station roles, 18 subsystems, 26 tools), qualifying the test system itself as a measurement system, not just the units passing it.
- Plan the test-software roadmap: sequence station bring-up against the production schedule, allocate engineering effort across workstreams, and own the timeline other teams plan against.
- Eliminated three classes of measurement error in the test software, a fixture-coupled magnetometer band, a saturating telemetry field and a boot-timing race, plus a threshold too wide to ever fail.
- Introduced mutation testing to prove the suite itself is robust, and restructured it into system code, test code and test profiles so a profile retunes on the line without a code release.
- Disproved an accepted data-classification method, only 37.8% correct, and rebuilt it from source.
- Derived test tolerances from measurement-system capability (gauge R&R), not convention.
- Added ratcheting coverage floors and a nightly adversarial code review over the test suite.
- Led a cross-program electromagnetic interference investigation and owned the corrective test gate.
- Built a hardware-in-the-loop captive-trajectory rig around a 6-axis force/torque sensor.

Microsoft Redmond, WA, USA

Senior Software Engineer Jan. 2022 - Mar. 2026

- Led test strategy for new Azure datacenter hardware, owning quality and manufacturability to volume.
- Drove coverage-versus-station-cost tradeoffs across the datacenter hardware portfolio.
- Turned ambiguous requirements into work instructions OEM/ODM operators executed reliably at scale.
- Supervised global OEM/ODM partners and built an operator certification program across the vendor base.
- Developed a mission-critical storage device driver in Rust for datacenter units.
- Built multi-agent autonomous systems keeping remote automated deployment stable at fleet scale.
- Reviewed team designs and code; root-caused issues across the Azure hardware/firmware/software stack.

Microsoft Pittsburgh, PA, USA

Senior Hardware Test Engineer Sep. 2020 - Dec. 2021

- Owned hardware test development for Azure Edge Computing with globally distributed OEM/ODM teams.
- Designed and integrated the automation test suite; defined hardware component test plans.
- Root-caused hardware and firmware defects across BMC, IPMI, BIOS, CPU, GPU, VPU, DIMM and NVMe.

Seagate Technology Minneapolis, MN, USA

Senior Firmware Engineer Feb. 2016 - Sep. 2020

- Primary developer of the logging and S.M.A.R.T. subsystems in a multi-CPU embedded system.
- Developed hard drive firmware in an RTOS environment on ARM Cortex microcontrollers.
- Applied Failure Mode and Effects Analysis (FMEA) across projects to surface reliability risk early.
- Worked across SAS, SATA, I2C, SPI, UART and JTAG interfaces; Agile/Scrum delivery with Git.
- Mentored new engineers onto the team and helped them ramp to independent ownership.

Bayer HealthCare New York, NY, USA

Device R&D Firmware Engineer, Internship May 2015 - Aug. 2015

- Built an Android app that auto-validated glucose meter firmware over BLE, cutting test time 94%.
- Cut the validation cycle from 64 hours/week to 4; built the non-invasive optical glucose test rig.

Intel Taipei, Taiwan

Firmware Validation Engineer Mar. 2013 - Jun. 2014

- Debugged firmware defects across twelve distinct Intel tablet platforms.
- Tested and certified fifteen-plus products against Microsoft WHCK and Google CTS.
- Root-caused defects across the software / operating system / hardware boundary.

Ambarella Hsinchu, Taiwan

DSP Firmware Developer Oct. 2011 - Jan. 2013

- Developed DSP firmware on an ARM11 dual Cortex platform under microITRON and ThreadX RTOS.
- Improved face and object tracking success rate by 40% and cut power use 4.2% on Samsung DV.
- Designed a real-time speed alert for CarDV using the Hough transform.
- Supported OEM and ODM on site to improve the manufacturing process.

Taiwan Air Force Taiwan

Communication Sergeant (Mandatory Military Service) Oct. 2010 - Sep. 2011

- Tracked unidentified aerial contacts on radar and relayed track data to position field artillery.
- Maintained encrypted wireless and wired equipment for secure transmission of classified information.

Freelance Programmer Taipei, Taiwan

Software and Firmware Developer Mar. 2009 - Sep. 2010

- Constructed an interactive installation for Taipei City Government (C# gesture detection, Arduino).
- Built an Arduino light control system for the Taiwan International Lighting Show, Taipei World Trade Center.

Education

The University of Texas at Dallas Dallas, TX, USA

Master of Science, Major in Computer Science, Data Science track. Dec. 2015

Honors & Awards

1st Prize, National Intelligent Robots Contest, Taiwan Ministry of Education Taipei, Taiwan

- Built a housekeeper robot from scratch, controlled remotely by a Nokia 3310.
- Analyzed user commands with a Dual Tone Multi Frequency (DTMF) receiver chip HT9170.
- News interview: http://youtu.be/sW_bO8OxeQI?t=34s

Skills

- **Manufacturing and Test:** line automation, test strategy, Design for Test (DFT), New Product Introduction (NPI), volume ramp, PCBA L5 ICT fixtures, BMC, IPMI, FMEA, gauge R&R, OEM/ODM management.
- **Software and Firmware:** Rust, C/C++, Python, embedded systems, device drivers, ARM, RTOS, Android.
- **Methods:** gauge R&R, measurement-system analysis, FMEA, mutation testing, root cause analysis.
- **Cloud and Tools:** Azure, AWS, Power BI, Git, Jira.
- **Languages:** native Traditional and Simplified Chinese; intermediate Japanese and Korean.
- **Clearance:** United States citizen, eligible to obtain a U.S. security clearance.